

IEC TS 60601-4-2 and EMC Guidance for Medical Devices

As medical devices grow more connected and complex, electromagnetic compatibility is under sharper scrutiny. IEC TS 60601-4-2 provides essential guidance for evaluating how non-medical functions perform under electromagnetic stress, ensuring devices remain reliable, safe, and effective in real-world healthcare environments.



Key Update: From Technical Report (TR) to Technical Specification (TS) – What It Really Means:

In the evolving landscape of medical EMC (Electromagnetic Compatibility) standards, a recent development may seem significant at first glance but is more evolutionary than revolutionary. We're referring to the transition of the IEC 60601-4-2 document from a Technical Report (TR) to a Technical Specification (TS).

At its core, this shift doesn't introduce sweeping technical changes. Instead, it reflects a natural maturation of the document. A Technical Report often serves as an early-stage resource, a sandbox of expert recommendations, best practices, and emerging insights. When a document progresses to a Technical Specification, it indicates that the content has become more stable, better understood, and more formally structured, while still retaining its status as guidance not a formal requirement.

What Stayed the Same?

First and foremost, both the TR and TS versions of IEC 60601-4-2 remain guidance documents. They are not enforceable standards in themselves, but they provide valuable insight into how manufacturers and test labs might approach the evaluation of non-medical functions within a medical electrical system or equipment, especially during immunity testing.

What's New?

While this isn't a technical overhaul, there are a few minor but noteworthy updates in the TS version:

- **Improved Alignment:** The TS now aligns more cleanly with the 4th and 4.1 editions of IEC 60601-1-2, the cornerstone EMC standard for medical electrical equipment. This harmonization reduces ambiguity and makes integration into test plans more streamlined.
- **Reference Cleanup:** The new TS features improved citations and cross-references, making it easier for users to locate related clauses, contextual standards, and applicable definitions.
- **Cable Length Exemption:** In the earlier TR, cables under 3 meters in length could be exempted from conducted RF immunity testing. The TS has tightened this guidance: now, only cables shorter than 1 meter may be exempted.

* Important Note: The U.S. FDA does not recognize this exemption. Regardless of cable length, the FDA expects all cables to be evaluated for conducted RF immunity. So, while international manufacturers might see some flexibility in global markets, those submitting to the FDA must plan to test every cable, regardless of length.

This TS update is less about redefining EMC expectations and more about refining how we articulate and apply them. It encourages

manufacturers to start incorporating these clarifications earlier in the design and risk management process, allowing for more coherent and justifiable test plans especially when preparing for regulatory submissions.

If you're navigating this transition or unsure how it may affect your product testing strategy, now is a good time to revisit your EMC test plans and documentation. These subtle changes can have ripple effects particularly if you're marketing in both the U.S. and international markets.

Immunity vs. Emissions – Understanding the Scope of TS 60601-4-2

To fully grasp the impact of TS 60601-4-2, it's important to understand what it covers and what it doesn't.

The focus of TS 60601-4-2 is exclusively on immunity. In other words, it's concerned with how well a medical electrical device can resist interference and function correctly in the presence of electromagnetic disturbances in its' intended use environment. Think Wi-Fi routers, microwaves, mobile phones—everyday sources of potential interference.



What about the electromagnetic emissions that the device itself might generate?

That's still the domain of IEC 60601-1-2, which governs both emissions and immunity. Emission limits are essential to ensure that one medical device doesn't interfere with another or with other collocated electronics. Manufacturers must take both into account. While TS 60601-4-2 helps you build resilient devices, you still need to consult IEC 60601-1-2 to ensure your device doesn't cause problems for others. It's important to remember that intended use environments matter. A device designed for professional clinical settings will face different electromagnetic conditions than one used in a private residence or nursing home. Devices must be engineered accordingly, or you risk poor performance or even safety issues.

Intended Use vs. Essential Performance – Why Both Matter in EMC Testing

When it comes to designing and testing medical devices, especially those intended for use in unpredictable home healthcare environments, it's crucial to draw a clear line between what a device is supposed to do and what it must do to keep patients safe. That's where the concepts of intended use, essential performance, and required performance come into play.

- Intended Use refers to the primary purpose and marketed function of the device. For example, if a wearable medical device is designed to collect and send diagnostic images to a hospital system, that data transmission is part of its intended use—even if the feature isn't critical to immediate patient safety.
- Essential Performance is narrower in scope. It covers the functions that are vital to patient safety that, if lost or degraded, could result in unacceptable risk. Think of a CT scanner's dosage control system: if that goes haywire, it could expose the patient to harmful levels of radiation. That's essential performance.
- Required Performance includes any functionality or behavior mandated by applicable particular standards. This might include minimum alert volume levels, alarm conditions, or uptime requirements that are specific to certain categories of devices.



Why This Matters in EMC Testing

Historically, EMC testing focused heavily on essential performance, sometimes to the exclusion of other functions. If a feature wasn't considered safety-critical, it might not have been evaluated under electromagnetic interference conditions. But TS 60601-4-2 changes that approach.

The updated specification makes it clear: manufacturers should evaluate all declared functions of the device under EMC conditions, not just those linked to essential performance. If your product's intended use includes features like wireless data transmission, remote monitoring, or app integration, those need to be tested for immunity and stability too, even if failure would not pose a direct risk to the patient. This is a key evolution, particularly in home environments where interference from consumer electronics is the norm, not the exception.

For regulators, healthcare professionals, and patients, a medical device must do more than just meet safety requirements. It has to work as promised. The push in TS 60601-4-2 to test for intended use alongside essential performance reflects this broader expectation: that devices will operate reliably, consistently, and as marketed even in real-world electromagnetic conditions.

As more devices move from clinical to consumer spaces, this shift becomes not just a good practice but an essential one.

Why Intertek

Intertek brings unmatched expertise in EMC testing for medical devices, combining deep knowledge of IEC 60601 standards with hands-on experience across the full spectrum of medical technologies. Our teams don't just run tests, we work in partnership with our clients, helping manufacturers anticipate regulatory expectations, reduce uncertainty, and accelerate time to market. With a service model built on responsiveness, technical insight, and global reach, we deliver confidence every day and with every project.

EMC testing is one part of Intertek's Total Quality Assurance approach. By integrating EMC with safety, performance, cybersecurity, and other assurance, testing, inspection, and certification services, we enable medical device manufacturers to design, certify, and launch products that meet the highest standards worldwide. From early development through global market access, Intertek is committed to ensuring devices are safe, reliable, and ready for patient care.

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Americas
+1 800 WORLDLAB (967 5352)
+1 251 459 6173

Europe
+44 1372 370900

Asia
+852 2173 8888



icenter@intertek.com



[Intertek.com/medical](https://www.intertek.com/medical)